Chapter Review

Key Terms

horizontal line A horizontal line is the graph of an equation that can be written in the form $y = b$. The line passes through the *y*-axis at $(0, b)$.

intercepts of a line Each of the points at which a line crosses the *x*-axis and the *y*-axis is called an intercept of the line.

linear equation An equation of the form $Ax + By = C$, where A and B are not both zero, is called a linear equation in two variables.

ordered pair An ordered pair (x, y) gives the coordinates of a point in a rectangular coordinate system. The first number is the x-coordinate. The second number is the y-coordinate.

$$\underset{x\text{-coordinate},y\text{-coordinate}}{(x,\ y)}$$

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origin The point $(0, 0)$ is called the origin. It is the point where the x-axis and y-axis intersect.

quadrants The x-axis and y-axis divide a rectangular coordinate system into four areas, called quadrants.

slope of a line The slope of a line is $m = \frac{\text{rise}}{\text{run}}$. The rise measures the vertical change and the run measures the horizontal change.

solution to a linear equation in two variables An ordered pair (x, y) is a solution to the linear equation $Ax + By = C$, if the equation is a true statement when the *x*- and *y*-values of the ordered pair are substituted into the equation.

vertical line A vertical line is the graph of an equation that can be written in the form $x = a$. The line passes through the *x*-axis at $(a, 0)$.

***x*-axis** The *x*-axis is the horizontal axis in a rectangular coordinate system.

***y*-axis** The *y*-axis is the vertical axis on a rectangular coordinate system.

Key Concepts

11.1 Use the Rectangular Coordinate System

- **Sign Patterns of the Quadrants**

Quadrant I	Quadrant II	Quadrant III	Quadrant IV
(*x*,*y*)	(*x*,*y*)	(*x*,*y*)	(*x*,*y*)
(+,+)	(−,+)	(−,−)	(+,−)

- **Coordinates of Zero**
  - Points with a *y*-coordinate equal to 0 are on the *x*-axis, and have coordinates (*a*, 0).
  - Points with a *x*-coordinate equal to 0 are on the *y*-axis, and have coordinates (0, *b*).
  - The point (0, 0) is called the origin. It is the point where the *x*-axis and *y*-axis intersect.

11.2 Graphing Linear Equations

- **Graph a linear equation by plotting points.**
  Step 1. Find three points whose coordinates are solutions to the equation. Organize them in a table.
  Step 2. Plot the points on a rectangular coordinate system. Check that the points line up. If they do not, carefully check your work.
  Step 3. Draw the line through the points. Extend the line to fill the grid and put arrows on both ends of the line.

- **Graph of a Linear Equation:**The graph of a linear equation $ax + by = c$ is a straight line.
  - Every point on the line is a solution of the equation.
  - Every solution of this equation is a point on this line.

11.3 Graphing with Intercepts

- **Intercepts**
  - The *x*-intercept is the point, $(a, 0)$, where the graph crosses the *x*-axis. The *x*-intercept occurs when y is zero.
  - The *y*-intercept is the point, $(0, b)$, where the graph crosses the *y*-axis. The *y*-intercept occurs when x is zero.
  - The *x*-intercept occurs when y is zero.
  - The *y*-intercept occurs when x is zero.

- **Find the *x* and *y* intercepts from the equation of a line**
  - To find the *x*-intercept of the line, let $y = 0$ and solve for *x*.